

STEP 1 — Install and Import Required Libraries

Step 1 — Install Stable Version

```
!pip install transformers==4.40.2 sentencepiece
```

```
Requirement already satisfied: transformers==4.40.2 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (from sentencepiece==0.1.99)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: huggingface-hub<1.0, >=0.19.3 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: tokenizers<0.20, >=0.19 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: hf-xet<2.0.0, >=1.1.3 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2)
Requirement already satisfied: charset-normalizer<4, >=2 in /usr/local/lib/python3.12/dist-packages (from requests==2.31.0)
Requirement already satisfied: idna<4, >=2.5 in /usr/local/lib/python3.12/dist-packages (from requests==2.31.0)
Requirement already satisfied: urllib3<3, >=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests==2.31.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests==2.31.0)
```

```
from transformers import pipeline

translator_fr = pipeline(
    "translation",
    model="Helsinki-NLP/opus-mt-en-fr"
)

print("Model loaded successfully")
```

```

che for model files in Transformers v4.22.0 has been updated. Migrating yo
0 [00:00<?, ?it/s]
ocal/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: Fu
ings.warn(
ocal/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: Userw
cret `HF_TOKEN` does not exist in your Colab secrets.
henticate with the Hugging Face Hub, create a token in your settings tab (
ll be able to reuse this secret in all of your notebooks.
note that authentication is recommended but still optional to access publ
ings.warn(
_model.bin: 100%                               301M/301M [00:02<00:00, 181MB/s]
ion_config.json: 100%                           293/293 [00:00<00:00, 26.8kB/s]
r_config.json: 100%                             42.0/42.0 [00:00<00:00, 3.43kB/s]
spm: 100%                                       778k/778k [00:00<00:00, 11.0MB/s]
pm: 100%                                       802k/802k [00:00<00:00, 47.9MB/s]
son:      1.34M/? [00:00<00:00, 46.0MB/s]
loaded successfully
ocal/lib/python3.12/dist-packages/transformers/models/marian/tokenization_
ings.warn("Recommended: pip install sacremoses.")

```

STEP 2 — Load Translation Model

```

from transformers import pipeline

translator_fr = pipeline(
    "translation",
    model="Helsinki-NLP/opus-mt-en-fr"
)

print("French translation model loaded successfully")

```

French translation model loaded successfully

STEP 3 — Prepare Input Sentences

```

sentences = [
    "Artificial intelligence is transforming technology.",
    "I am learning natural language processing.",
    "The weather is beautiful today.",
    "She completed the project successfully.",
    "The teacher explained the concept clearly.",
    "John went to the market and bought fruits.",
    "Although it was raining, they played football.",
    "If you work hard, you will achieve success.",
    "Why are you late today?",
    "Did you finish your homework?",
    "Machine learning models require large datasets.",

```

```
"The research paper discusses deep learning techniques.",  
"Social media influences public opinion.",  
"Students are preparing for their examinations.",  
"Technology is changing the way we communicate."  
]
```

STEP 4 — Perform Translation (English → French)

```
print("English → French Translation\n")  
  
for sentence in sentences:  
    result = translator_fr(sentence)[0]['translation_text']  
    print("English:", sentence)  
    print("French :", result)  
    print()
```

English → French Translation

English: Artificial intelligence is transforming technology.
French : L'intelligence artificielle transforme la technologie.

English: I am learning natural language processing.
French : J'apprends le traitement naturel du langage.

English: The weather is beautiful today.
French : Le temps est beau aujourd'hui.

English: She completed the project successfully.
French : Elle a terminé le projet avec succès.

English: The teacher explained the concept clearly.
French : L'enseignant a expliqué clairement le concept.

English: John went to the market and bought fruits.
French : John est allé au marché et a acheté des fruits.

English: Although it was raining, they played football.
French : Même s'il pleuvait, ils jouaient au football.

English: If you work hard, you will achieve success.
French : Si vous travaillez dur, vous réussirez.

English: Why are you late today?
French : Pourquoi es-tu en retard aujourd'hui ?

English: Did you finish your homework?
French : Tu as fini tes devoirs ?

English: Machine learning models require large datasets.
French : Les modèles d'apprentissage automatique nécessitent de gros ensembles

English: The research paper discusses deep learning techniques.
French : Le document de recherche traite des techniques d'apprentissage profond

English: Social media influences public opinion.
French : Les médias sociaux influencent l'opinion publique.

English: Students are preparing for their examinations.

French : Les étudiants se préparent à leurs examens.

English: Technology is changing the way we communicate.

French : La technologie change la façon dont nous communiquons.

STEP 5 — Translate into Another Language (German)

```
translator_de = pipeline(  
    "translation",  
    model="Helsinki-NLP/opus-mt-en-de"  
)  
  
print("English → German Translation\n")  
  
for sentence in sentences:  
    result = translator_de(sentence)[0]['translation_text']  
    print("English:", sentence)  
    print("German :", result)  
    print()
```



```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: F
warnings.warn(

config.json:      1.33k/? [00:00<00:00, 63.1kB/s]

pytorch_model.bin: 100%                               298M/298M [00:04<00:00, 95.1MB/s]

generation_config.json: 100%                           293/293 [00:00<00:00, 19.7kB/s]
STEP 6 — Analyze Translation Quality
tokenizer_config.json: 100%                             42.0/42.0 [00:00<00:00, 3.78kB/s]

```

The translation results generated by the pretrained transformer models were generally accurate and fluent. Most simple sentences were translated correctly while preserving the original meaning. Compound and complex sentences sometimes showed slight changes in word order due to grammatical differences between languages. Technical terms related to technology and machine learning were mostly translated correctly. The English → German Translation

English: Artificial intelligence is transforming technology.
German: Künstliche Intelligenz transformiert die Technologie.

English: Why do people rely on technology for daily communication?
German: Warum verlassen sich die Menschen auf Technologie für die tägliche Kommunikation?

English: The research paper explains how machine learning models analyze large datasets.
German: Das Forschungspapier erklärt, wie Modelle des maschinellen Lernens große Datensätze analysieren.

English: Even though the project was challenging, the team completed it successfully.
German: Obwohl das Projekt herausfordernd war, hat das Team es erfolgreich abgeschlossen.

Overall, the pretrained models provided high-quality translations without requiring any training. However, automatic translation systems may struggle with idiomatic expressions, cultural context, and specialized domain vocabulary.

English: The weather is beautiful today.
German: Das Wetter ist heute schön.

STEP 7 — Test with Complex Sentences

English: She completed the project successfully.
German: Sie hat das Projekt erfolgreich abgeschlossen.

```

complex_sentences = [
    "Although artificial intelligence is powerful, it must be used responsibly.",
    "If students practice coding regularly, they will improve their programming sk",
    "The research paper explains how machine learning models analyze large dataset",
    "Why do people rely on technology for daily communication?",
    "Even though the project was challenging, the team completed it successfully."
]

print("Complex Sentence Translation\n")

for s in complex_sentences:
    result = translator_fr(s)[0]['translation_text']
    print("English:", s)
    print("French :", result)
    print()

```

Complex Sentence Translation

English: Machine learning models require large datasets.
German: Maschinelle Lernmodelle erfordern große Datensätze.

English: Although artificial intelligence is powerful, it must be used responsi
French: Bien que l'intelligence artificielle soit puissante, elle doit être ut

English: The research paper discusses deep learning techniques.
German: In dem Forschungspapier werden Techniken des tiefen Lernens diskutiert

English: If students practice coding regularly, they will improve their program
French: Si les étudiants pratiquent le codage régulièrement, ils amélioreront

English: Social media influences public opinion.
German: Soziale Medien beeinflussen die öffentliche Meinung.

English: The research paper explains how machine learning models analyze large
French: Le document de recherche explique comment les modèles d'apprentissage

English: Students are preparing for their examinations.
German: Die Studierenden bereiten sich auf ihre Prüfungen vor.

English: Why do people rely on technology for daily communication?

English: Technology is changing the way we communicate.
 French: Pourquoi les gens comptent-ils sur la technologie pour la communication?
 German: Technologie verändert die Art und Weise, wie wir kommunizieren.

English: Even though the project was challenging, the team completed it successfully.
 French: Même si le projet était difficile, l'équipe l'a mené à bien.

```
test_sentences = [
    "The movie was very interesting.",
    "Students are learning machine learning.",
    "Technology is improving rapidly."
]

for s in test_sentences:
    result = translator_fr(s)[0]['translation_text']
    print("English:", s)
    print("French :", result)
    print()
```

English: The movie was very interesting.
 French : Le film était très intéressant.

English: Students are learning machine learning.
 French : Les étudiants apprennent l'apprentissage automatique.

English: Technology is improving rapidly.
 French : La technologie s'améliore rapidement.

```
test_sentences = [
    "Artificial intelligence is changing the world.",
    "Why are you studying natural language processing?",
    "If you practice coding, you will become better.",
    "Although it was raining, they went outside."
]

for s in test_sentences:
    result = translator_fr(s)[0]['translation_text']
    print("English:", s)
    print("French :", result)
    print()
```

English: Artificial intelligence is changing the world.
 French : L'intelligence artificielle change le monde.

English: Why are you studying natural language processing?
 French : Pourquoi étudiez-vous le traitement du langage naturel?

English: If you practice coding, you will become better.
 French : Si vous pratiquez le codage, vous vous améliorerez.